

Anuj Kamath

+91 9013469195 Udupi, Karnataka anujkamath1@gmail.com
github.com/anujkamath linkedin.com/anujkamath

SUMMARY

Computer Science and Engineering student with hands-on experience in training deep learning models for tasks in natural language and speech processing. Experienced in implementing attention-based architectures and evaluation pipelines. Also familiar with end-to-end application development, contributing to both backend logic and frontend integrations in real-time systems. Interested in solving problems at the intersection of machine learning and scalable software systems.

EDUCATION

BTech in Computer Science and Engineering, Manipal Institute of Technology, Manipal Expected 2026

- Enrolled in the **Honors Program**, working on advanced topics in **Natural Language Processing (NLP)**.
- Completed **PwC Data Analytics and AI Launchpad** (Professional Certification).

EXPERIENCE

Software Engineer Intern June 2024 - August 2024
Emper.ai Hyderabad, Telangana

- Built backend services using Node.js and Express, supporting core application features.
- Created interactive dashboards for real-time data visualization using React.js and Tailwind CSS.
- Integrated third-party APIs to enable authentication and many other functionality.
- Designed and implemented role-based access control to improve application security.
- Collaborated with the team on database query optimization to support efficient data access.

Engineer Intern July 2024 - July 2024
WiseStep Hyderabad, Telangana

- Developed REST APIs for user and session management, contributing to backend functionality improvements.
- Assisted in optimizing SQL queries to enhance data retrieval efficiency for analytics tasks.
- Fixed bugs in existing Node.js microservices, improving application stability.
- Participated in code refactoring to simplify workflows and reduce execution time.
- Conducted performance and load testing to support reliable system operations.

Chairperson Jul 2024 - Jul 2025
IECSE Manipal, Karnataka

- Organized multiple coding competitions, managing logistics and participant engagement.
- Led a team of over 50 members to successfully execute large-scale technical events on time.
- Coordinated with industry professionals to arrange guest lectures and mentorship sessions.

PROJECTS

- Receipt Data Extraction using OCR** - Built a pipeline using OpenCV and Tesseract OCR to extract and structure data from receipts. Applied preprocessing techniques like Gaussian blurring and adaptive thresholding to improve OCR accuracy by 40%. Integrated Named Entity Recognition (NER) to categorize extracted data into structured fields.
- Applicant Tracking System (Hyrd)** - Developed a MERN-stack-based ATS with resume parsing, advanced filtering, and querying capabilities, enabling HR teams to process large applicant data efficiently. Executed a job recommendation engine using TF-IDF and cosine similarity for matching candidates with suitable roles. Optimized MongoDB queries, reducing response times by 35.
- Emotion Recognition in Speech** - Designed a deep learning model for speech-based emotion classification using MFCC feature extraction and a CNN-RNN architecture. Fine-tuned hyperparameters using grid search, achieving a classification accuracy of 87%. Built a real-time demo interface for visualizing emotion trends in audio recordings.
- Speech-to-Speech with Prosody and Sentiment Analysis** - Developed a pipeline that converts speech while preserving prosody and sentiment. Used prosodic feature extraction and sentiment-aware transformations to ensure natural, expressive speech output. Implemented a multi-stage pipeline for synthesis and sentiment-driven modifications.

SKILLS AND COMPETENCIES

Programming Languages: Python, C/C++, Java

Machine Learning & NLP: Supervised and unsupervised learning, model training and evaluation, feature extraction, data preprocessing, NLP pipelines, text classification, sentiment analysis, speech processing, Transformer-based models, attention mechanisms

ML & NLP Libraries / Frameworks: PyTorch, TensorFlow, scikit-learn, NLTK, NumPy, Pandas

Backend & Systems (Supporting): REST APIs, Node.js, Express.js, Django, basic understanding of distributed systems

Databases: PostgreSQL, MySQL, MongoDB

Web Development (Supporting): JavaScript, React.js, HTML, CSS, Tailwind, Next.js, Docker, Azure, AWS

Computer Science Fundamentals: Data structures and algorithms, machine learning fundamentals, operating systems

ACHIEVEMENTS

- Branch Topper (09/2022 - 06/2023) - First Year CGPA: 9.74/10